

INSTRUCTOR Solution - FINAL EXAM Sp26

[15 pts]

Q1) (a)

[3 pt]

$$P(D) = \sum_i P(G_i) \cdot P(D|G_i) \quad \text{--- (1)}$$

(total probability theorem)

$$\begin{aligned} &= P(L) \cdot P(D|L) + P(M) \cdot P(D|M) + P(H) \cdot P(D|H) \\ &= (0.7 \cdot 0.01) + (0.2 \cdot 0.05) + (0.1 \cdot 0.2) \quad \text{--- (1)} \\ &= 0.007 + 0.01 + 0.02 = 0.037 \quad \text{--- (1)} \end{aligned}$$

$$P(D) = 0.037 = 3.7\%$$

(b) $P(+ve) = P(+ve|D) \cdot P(D) + P(+ve|D^c) \cdot P(D^c)$ --- (1)

$$\begin{aligned} &= (0.95 \cdot 0.037) + (0.08 \cdot (1 - 0.037)) \\ &= 0.03515 + 0.07704 \quad \text{--- (1)} \\ &= 0.11219 \quad \text{--- (1)} \end{aligned}$$

[3 pts]

(c) $P(D|+ve) = \frac{P(+ve|D) \cdot P(D)}{P(+ve)}$ --- (1)

using Bayes Rule

$$= \frac{0.03515}{0.11219} = 0.3133 \quad \text{--- (1)}$$

[4 pts]

(d) $P(H|+ve) = \frac{P(+ve|H) \cdot P(H)}{P(+ve)}$ --- (1)

[5 pts]

lets find

$\Rightarrow$

$$\begin{aligned} P(+ve|H) &= P(+ve|D \cap H) \cdot P(D|H) + P(+ve|D^c \cap H) \cdot P(D^c|H) \\ &= (0.95 \cdot 0.2) + (0.08 \times (1 - 0.2)) \\ &= 0.19 + 0.064 = 0.254 \quad \text{--- (2)} \end{aligned}$$

$$P(H|+ve) = \frac{0.254 \cdot 0.1}{0.11219} = 0.2264 \quad \text{--- (1)}$$

Q2

(a) $X \sim \text{Binomial}(n=2, p=1/2)$

[5pt]

Since, $X' = 2X - 2$
 $\Rightarrow P(X=0) = 1/4, P(X=1) = 1/2 \text{ \& } P(X=2) = 1/4$
 $\Rightarrow P(X'=-2) = 1/4, P(X'=0) = 1/2 \text{ \& } P(X'=2) = 1/4$

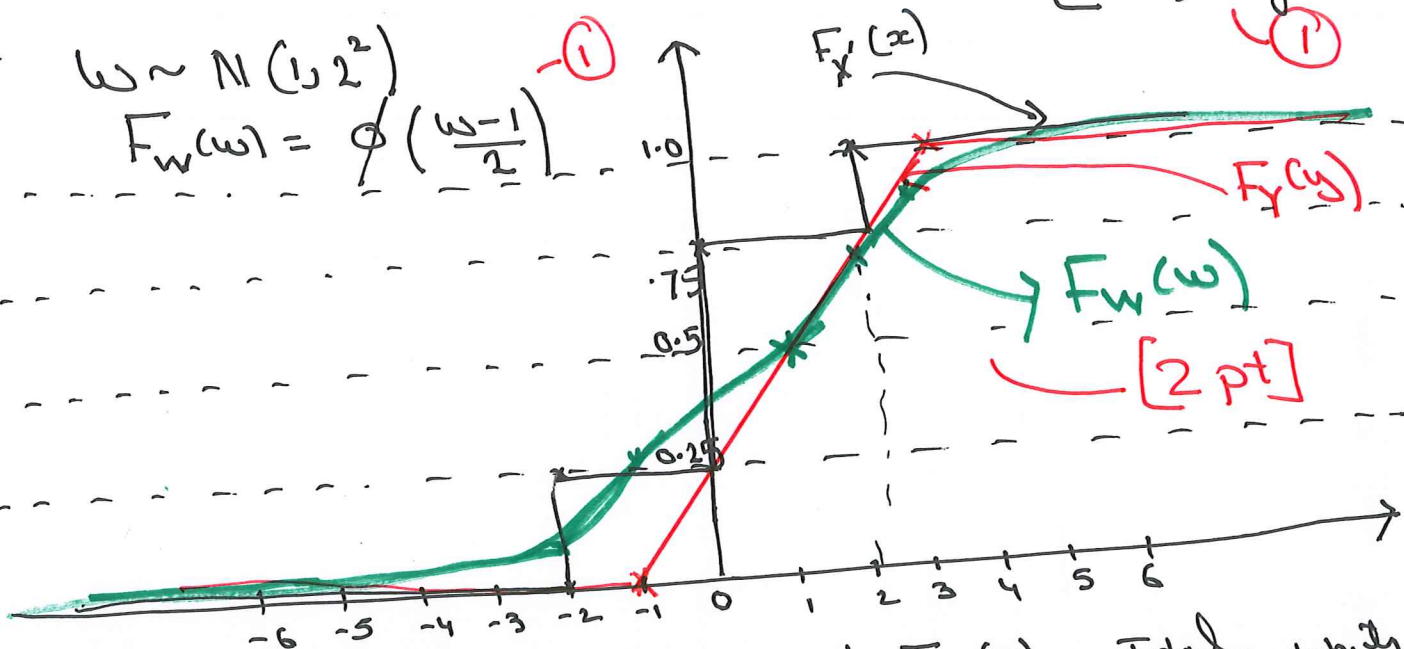
$$F_{X'}(x) = \begin{cases} 0, & x' < -2 \\ 1/4, & -2 \leq x' < 0 \\ 3/4, & 0 \leq x' < 2 \\ 1, & 2 \leq x' \end{cases} \quad (1)$$

for $Y \sim U(-1, 3)$

$$F_Y(y) = \begin{cases} 0, & y < -1 \\ \frac{y+1}{4}, & -1 \leq y \leq 3 \\ 1, & y > 3 \end{cases} \quad (1)$$

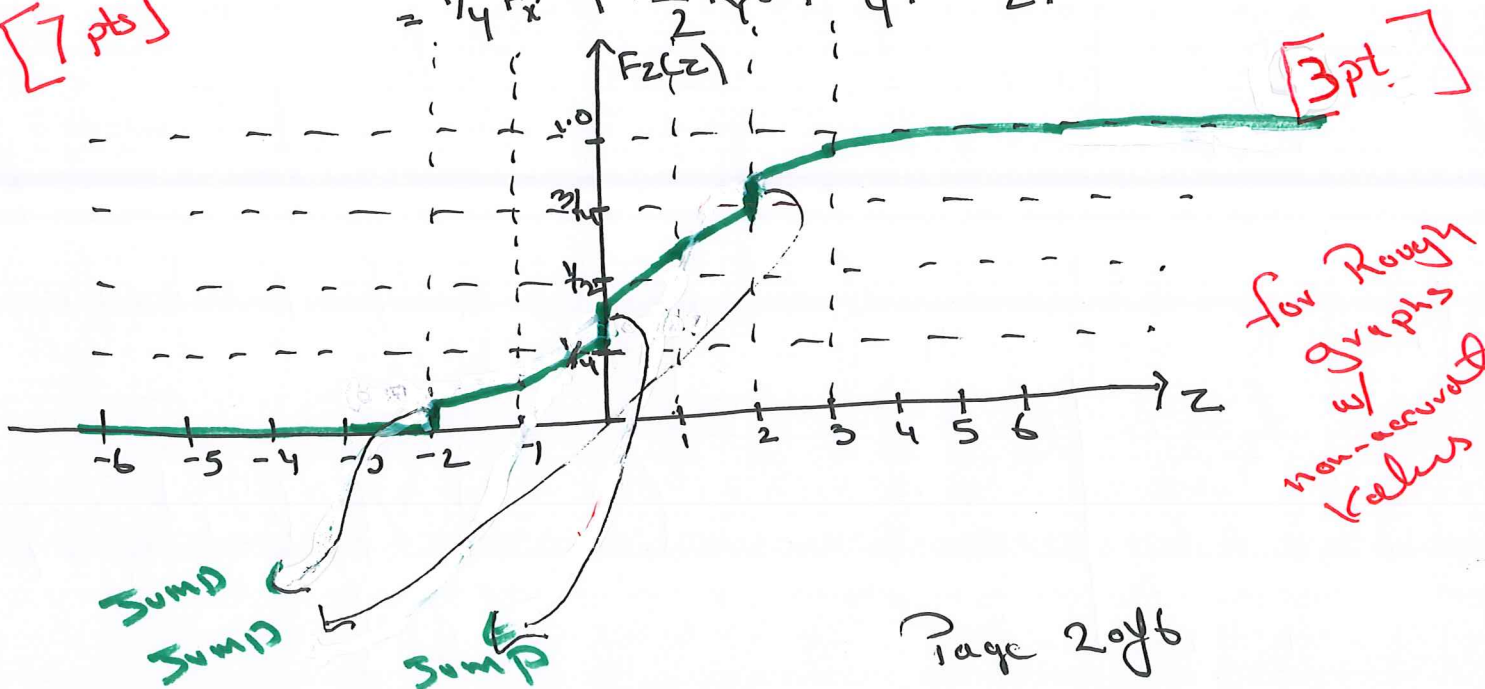
for $W \sim N(1, 2^2)$

$$F_W(w) = \Phi\left(\frac{w-1}{2}\right)$$

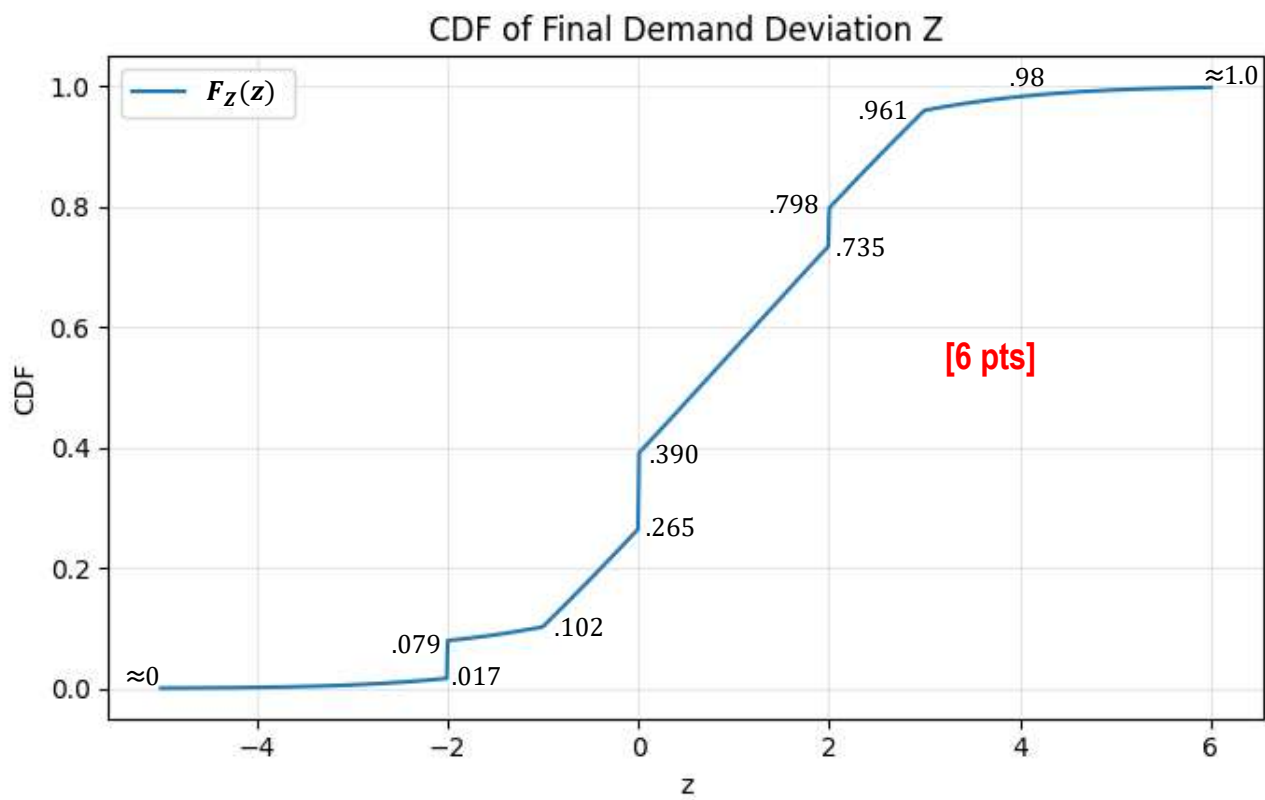


(b) $F_Z(z) = \frac{1}{4} F_{X'}(z) + \frac{1}{2} F_Y(z) + \frac{1}{4} F_W(z)$ - Total probability theorem
 $= \frac{1}{4} F_{X'}(z) + \frac{1}{2} F_Y(z) + \frac{1}{4} \Phi\left(\frac{z-1}{2}\right)$ [1 pt]

[7 pts]



for Rough graphs w/ non-accurate values,



z	$F_Z(z-)$ <i>(before jump value)</i>	$F_Z(z)$	
-infinity		0	
-2	0.017	0.079	jump
-1		0.102	
0	0.265	0.390	jump
1		0.563	
2	0.735	0.798	jump
3		0.961	
+infinity		1	

for Calculations

Z	Prob x' 1/4	Y 1/2	W 1/4	F _Z (z) ↓ Probab. ↓
-2	0 0.25	0	0.067	0.016 Sum ↑ 0.07925
-1	0.25	0	0.1587	0.102
0	0.25 0.75	0.25	0.308	0.264 Sum ↑ 0.389
1	0.75	0.5	0.5	0.5625
2	0.75 1	0.75	0.69	0.735 Sum ↑ 0.7975
3	1	1	0.8413	0.96
4	1	1	0.933	0.98

(c) $P(-1 < Z \leq 2)$

[3 pt]

$F_Z(2) = 0.7975$

1 pt

$F_Z(-1) = 0.102$

1 pt

$P(-1 < Z \leq 2) = 0.695$

1 pt

Q3)

(a)

$$P[60 \leq X \leq 75]$$

Can be open or closed interval

[4 pts]

$$= P\left[\frac{60-68}{12} \leq Y \leq \frac{75-68}{12}\right]$$

where $X \sim N(68, 12^2)$ where $Y \sim N(0, 1)$

$$= P(-0.666 \leq Y \leq 0.583) \text{ --- [1 pt]}$$

$$= P(Y \leq 0.583) - P(Y \leq -0.666) \text{ --- [1 pt]}$$

$$= 1 - P(Y \leq -0.666)$$

Using Normal Table

$$= 0.71904 - (1 - 0.748) \text{ --- [1 pt]}$$

$$= 0.71904 - 0.252 = 0.467 \text{ --- [1 pt]}$$

[6 pts]

$$(b) P(X \geq 80 | X \geq 50) = \frac{P(X \geq 80 \cap X \geq 50)}{P(X \geq 50)} \text{ [1 pt]}$$

$$X \sim N(68, 12^2)$$

$$Y \sim N(0, 1)$$

$$= \frac{P(X \geq 80)}{P(X \geq 50)} = \frac{1 - P(X < 80)}{1 - P(X < 50)} \text{ [1 pt]}$$

$$= \frac{1 - P(Y \leq \frac{80-68}{12})}{1 - P(Y \leq \frac{50-68}{12})} = \frac{1 - P(Y \leq 1)}{P(Y \leq -1.5)} \text{ [2 pt]}$$

[1 pt] Using Normal Table

of the students Passed, 17% have merit

$$= \frac{1 - 0.8413}{0.9332} = \frac{0.1587}{0.9332} = 0.17 \text{ [1 pt]}$$

(c)

Requirement

$$P(X \geq M) = 0.08 \Rightarrow P(X < M) = 0.92 \text{ [2 pt]}$$

$$\text{for a Normal Standard } P(Y < m') = 0.92 \text{ --- [1 pt]}$$

$$m' \approx 1.41 \text{ --- [1 pt]}$$

$$\Rightarrow \frac{M - 68}{12} \approx 1.41 \Rightarrow M \approx 84.92 \text{ or } 85\%$$

[1 pt]

[1 pt]

Q4) (a) $P(X \geq 9 | X \geq 1) = \frac{P(X \geq 9 \text{ and } X \geq 1)}{P(X \geq 1)}$ — (2)

Since $P(X > t) = \frac{2}{2+t}$ (for $t > 0$) $\Rightarrow$ $\frac{P(X \geq 9)}{P(X \geq 1)} = \frac{\frac{2}{2+9}}{\frac{2}{2+1}} = \frac{\frac{2}{11}}{\frac{2}{3}} = \frac{3}{11}$ — (1)

$\Rightarrow P(X \leq 9 | X \geq 1) = 1 - \frac{3}{11} = \frac{8}{11} = 0.727$ — (1) $= \frac{8}{11}$

$P(\text{at least one bulb is working})$
 $= 1 - \left(\frac{8}{11}\right)^3$ — (2)
 $= 1 - 0.384$
 $= 0.615$ — (1)

[7pts] (b) $P(Y \leq y) = P[\max\{X_1, X_2, \dots, X_n\} \leq y]$ — (1)

$= P[(X_1 \leq y) \cap (X_2 \leq y) \cap \dots \cap (X_n \leq y)]$ — (2)

Since, $f_X(x) = \begin{cases} \frac{1}{a} & 0 < x < a \\ 0 & \text{else} \end{cases} \Rightarrow P[X_1 \leq y] \cdot P[X_2 \leq y] \cdot \dots \cdot P[X_n \leq y]$ — (1)

$= \left(\frac{y}{a}\right) \cdot \left(\frac{y}{a}\right) \cdot \dots \cdot \left(\frac{y}{a}\right)$ — (2)

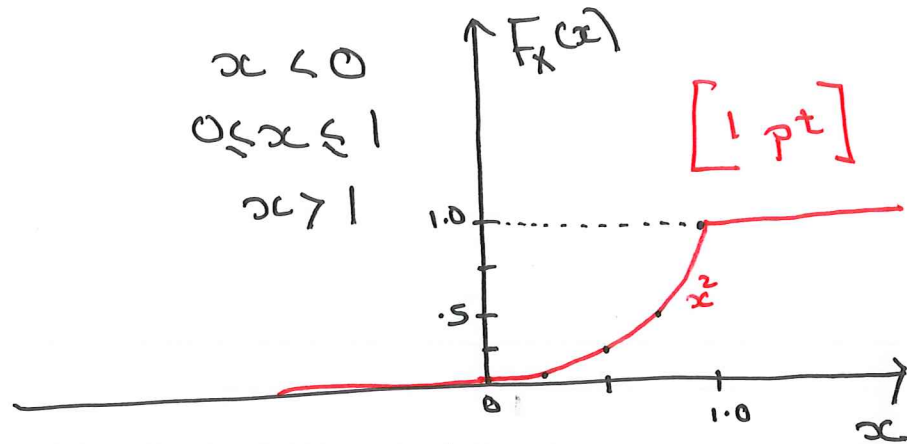
$P(Y \leq y) = \left(\frac{y}{a}\right)^n$ — (1)

Question 5 [15 pts]

(a) $F_X(x) = P(X \leq x) = \int_0^x f_X(x) dx$, $f_X(x) = 2x$ $0 \leq x \leq 1$
 0 , else
 [3 pts]

$\Rightarrow F_X(x) = x^2$ $0 \leq x \leq 1$ [1 pt]

$F_X(x) = \begin{cases} 0 & x < 0 \\ x^2 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$ [1 pt]



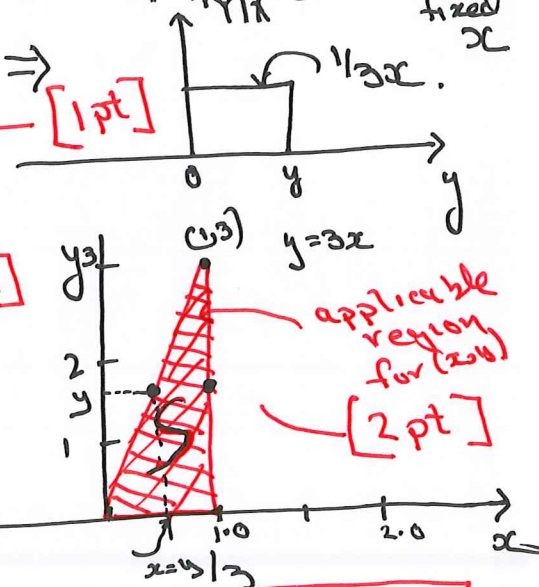
(b) $f_{X,Y}(x,y) = f_X(x) \cdot f_{Y|X}(y|x)$
 [6 pts] $= 2x \cdot f_{Y|X}(y|x)$ — [1 pt]

$Y|X \sim U(0, 3x)$ for fixed x

So, $f_{X,Y}(x,y) = 2x \times \frac{1}{3x}$, $0 \leq x \leq 1$
 $0 \leq y \leq 3x$ — [1 pt]
 $= 2/3$

$f_{X,Y} = \begin{cases} 2/3 & 0 \leq x \leq 1 \\ & 0 \leq y \leq 3x \\ 0 & \text{else} \end{cases}$ — [2 pt]

Uniform on S



(c) $f_Y(y) = \int_{x=y/3}^1 f_{X,Y}(x,y) dx$, when $Y=y$,
 $= \int_{y/3}^1 2/3 dx = 2/3 x \Big|_{y/3}^1 = 2/3 (1 - y/3)$ [1 pt]

x goes from $y/3$ to 1 [1 pt]

$f_Y(y) = \begin{cases} 2/3 (1 - y/3) & 0 \leq y \leq 3 \\ 0 & \text{else} \end{cases}$ [2 pt]

